



Internship Research Report

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Impacts of ENSO phenomena on skipjack tuna fisheries, and bigeye tuna fishing ground in relation to thermocline in the Western and Central Pacific Ocean

Abstract: It is well established now that the zonal displacement of the warm waters of the western equatorial Pacific occurring during the development of an El Niño or La Niña event has a major impact on the distribution of the surface tuna stocks^[1]. In this paper we found that the distribution of the abundance index CPUE (Catch Per Unit Effort) shifts longitudinally in concert with ENSO events distributed eastwards during El Niño phases, and westwards during La Niña phases, which shows an important relationship between the ENSO phenomena and fisheries. The relationship between thermocline and bigeye tuna (*Thunnus obesus*) fishing grounds in the Western and Central Pacific Ocean was evaluated by Argo data (BOA_Argo global Marine Argo grid data sets) and CPUE (catch per unit effort). The generalized additive model indicated evidence of nonlinear relationships between CPUE and four thermocline characteristics. The results suggested that the fishing grounds distributed where the upper boundary temperature was about 28 °C and the upper boundary depth values between 50 and 100 m. The fishing grounds located between the two high value shapes of the lower boundary depth of thermocline, if the depth was >400 or <250 m, the CPUE tended to be low.

Key words: Bigeye; Skipfish; ENSO; Thermocline parameters; Argo data sets

1. Introduction

Recently, some important inter-annual fluctuations have been observed in the fisheries, more or less in apparent correlation with the ENSO events as indicated by the ONI (Fig 1), we can learn that an El Niño is frequently followed or preceded by a La Niña with a frequency oscillating between 3 to 6 years. Therefore, it is necessary to consider long time-series of data for investigating cross-correlation or spectral analyses, all the more since responses to potential changes related to ENSO can be delayed through the repercussions in the population structure. However, it is often difficult to clearly demonstrate such correlation because several causes can be responsible for observed inter-annual fluctuations. ^[2] From a general point of view, to answer the question why the abundance of tuna species has changed in a given time and space, we have to consider three categories of factors: factors related to changes in fishing techniques that cause changes in species catch-ability (e.g., changing the depth exploited by the long-line), environmental factors inducing spatial changes in the distribution and movements of fish, both in the vertical and horizontal dimensions (e.g., in relation to the depth of the thermocline, or the seasonal or ENSO-related extension of warm waters), and finally real changes in abundance of the stock, with low or high levels of recruitment, in relation either to an environmental change or to the size of the spawning stock biomass (stock-recruitment relationship).

In order to figuring out the impacts other physical parameter, such as current's speed, wind's speed, temperature, thermocline, salinity ect. we adopted Argo profile buoys data, evaluated the isoline distribution of thermocline characteristics, calculated the optimum ranges of thermocline characteristics of the central fishing grounds, all of these contents will give some references to the tuna resource management in WCPO.

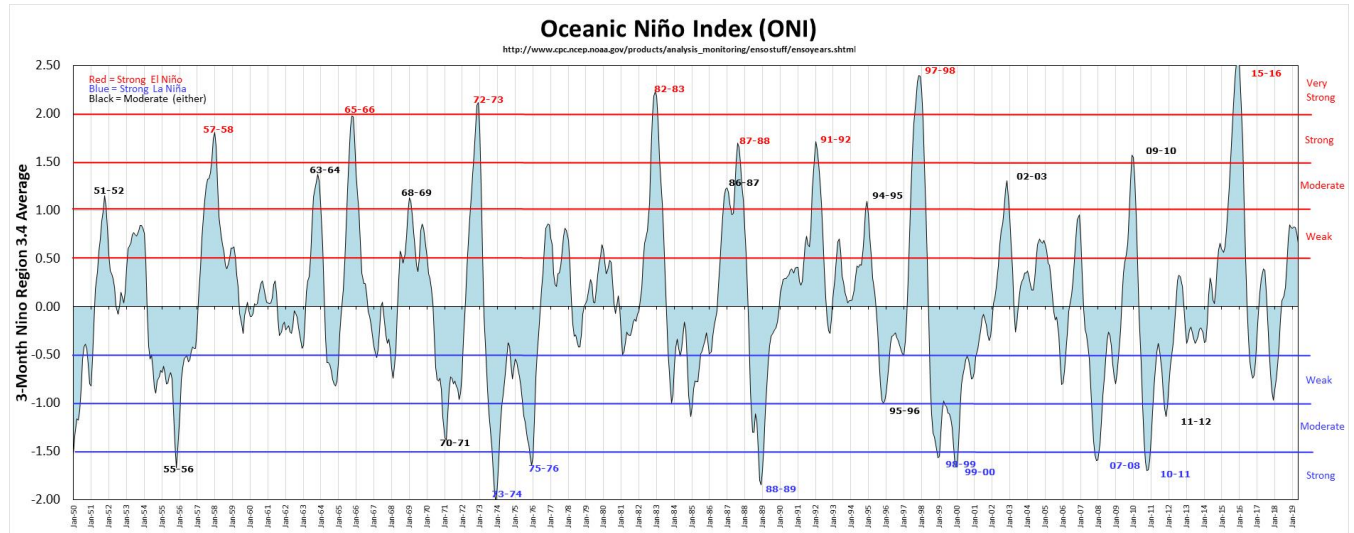


Fig 1 ONI Index, showing the annual changes of ENSO phenomena

2. ENSO phenomena

With the development of the fishery industry, the instruments have been rapidly evolved in recent 50 years, which will be favorable to increase the efforts to catch fishes. The purse seine fisheries in the tropical Pacific Ocean target mainly skipjack and yellow-fin. In this paper, the research target is skipjack tuna, and we use the relative fishery data downloaded at the WCPFC official website (<https://www.wcpfc.int/>).

As we can see the changes in number of invested purse seines and caught skipjack tuna (Fig 2a) .In recent 40 years, the catch efforts becomes stronger, while the CPUE gradually decreases. The high CPUE positions are moving northward and southward respectively, because of the global climate change, the rising of the sea temperature lead to the migration of the skipjack tuna (Fig 2b). This kind of new phenomenon can be come down to both naturally and artificially causes that we need pay more attention on this side. In contrast the ENSO's effect on the fisheries is much more regular and obvious, we can make some simple conclusions upon that.

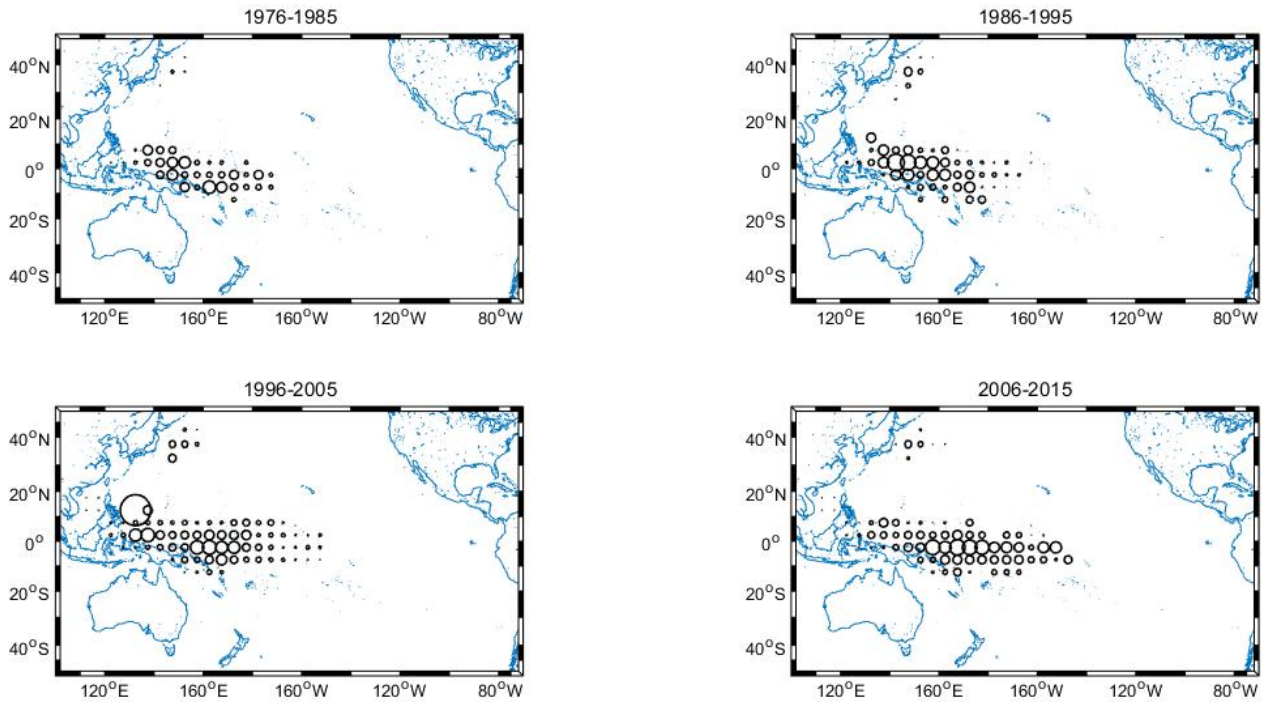


Fig 2a Transition of fishery distribution in Western and Central Ocean (1976-2015), the size of circle presents the amount of catching in one unit grid area ($5^{\circ}\times 5^{\circ}$)

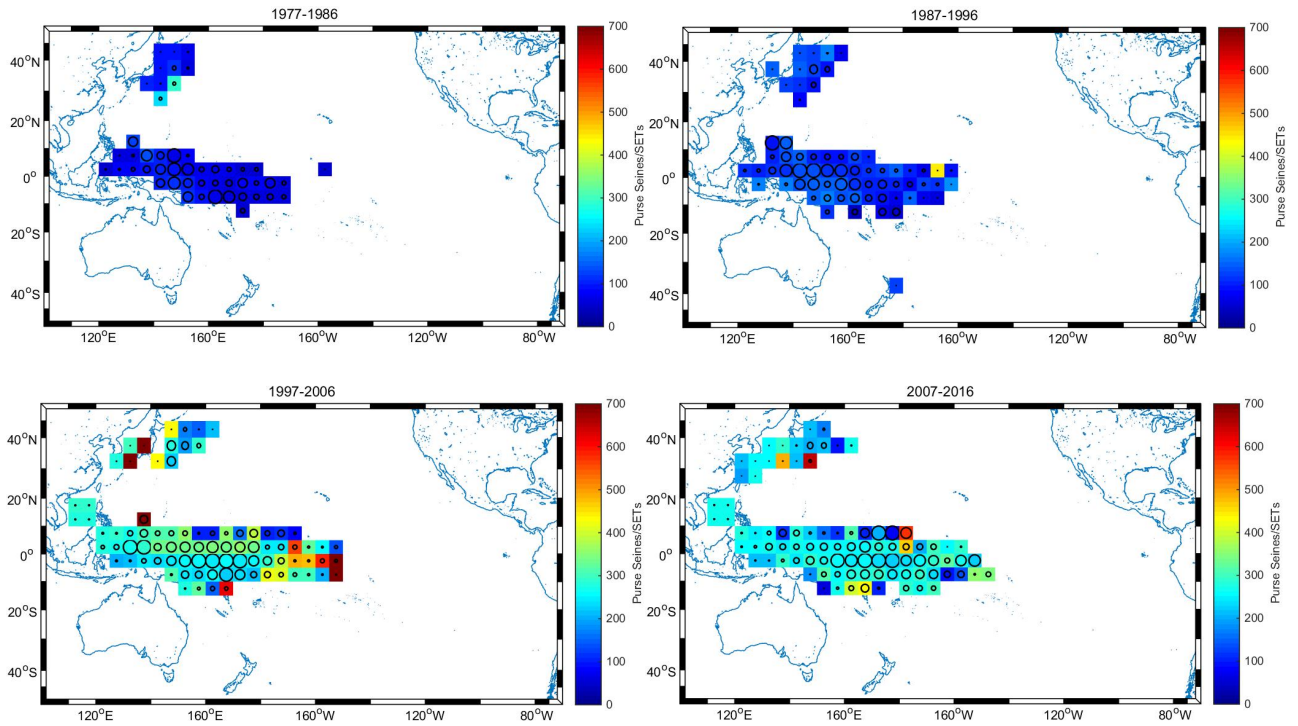


Fig 2b Transition of fishery distribution in Western and Central Pacific Ocean (1977-2016, overlapped the catching effort), the size of circle presents the amount of catching, while the Colorbar shows the average numbers of purses seines used in one unit grid area ($5^{\circ}\times 5^{\circ}$)

ENSO is an irregular low-frequency oscillation between a warm (El Niño) and cold (La Niña) state, that evolves under the influence of the dynamic interaction between atmosphere and ocean. The warm waters in the surface layer of the western equatorial Pacific (the warm pool) have a temperature above 28°C year-round, inducing an atmospheric convection connected to the colder eastern Pacific Ocean through the atmospheric

Walker circulation.

In the following analyses, special attention to spatial effects will be given for the most powerful El Niño events of 1997 and 2015 and the preceding or following La Niña events of 1989 and 2011 by examining the data of these respective years, we also used two normal years' temperature distribution data for comparison. The skipjack fishery data were obtained on the website of WCPFC and the high resolution global daily temperature abnormal datasets were downloaded on the website of NOAA (National Oceanic Atmospheric Administration), (<https://www.esrl.noaa.gov/psd/data/gridded/data.noaa.oisst.v2.highres.html>).

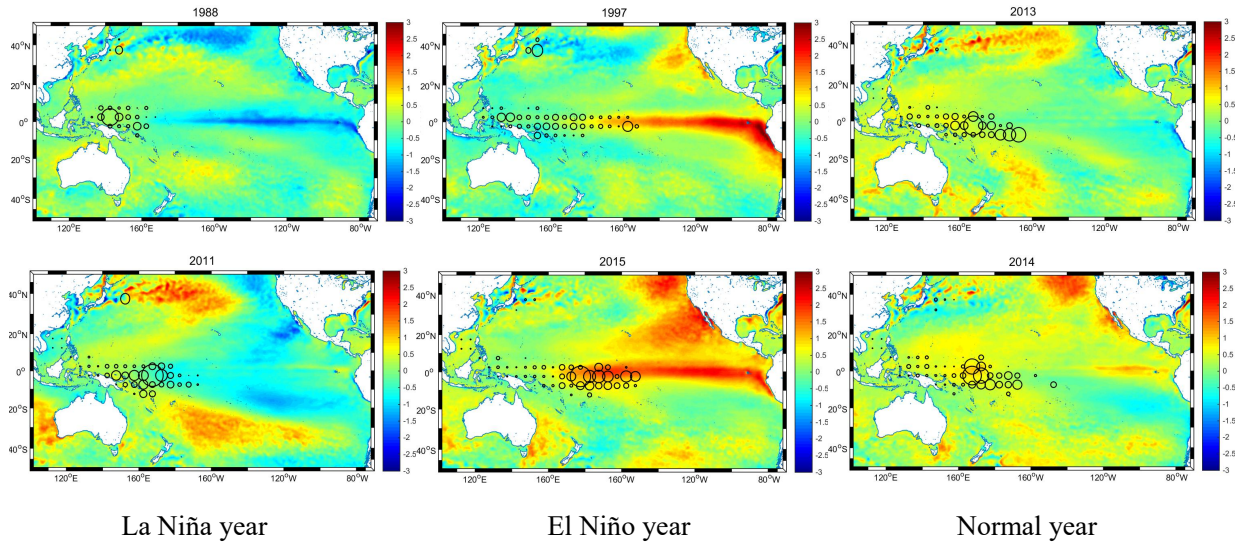


Fig 3 The distribution of fishery catch in western and central ocean under the effect of ENSO, the size of circle presents the amount of catching in one grid unit ($5^{\circ} \times 5^{\circ}$), while the Colorbar shows abnormal temperature

Spatial distributions of purse seine catch during ENSO events are presented in Figure 3. Although the fishery was not yet well developed in the WCPO during the 1997-2000 ENSO episodes, the eastward spatial shift during El Niño in this region is already visible. In contrast, during the La Niña year, we can notice the western spatial shift. In conclusion, ONI is a significant index (Fig.1) to inform us how the distribution of skipjack tuna would change in the WCPO during the ENSO period.

3. Thermocline parameters

3.1 Research target

Because of the habitat of the pelagic fish, it depends largely on the vertical structure of the water temperature. Many fish have the habit of moving vertically around the clock, and the thermocline acts like a natural barrier, affecting the up and down movement and living habits of fish. Bigeye tuna (*Thunnus obesus*) inhabit tropical and temperate waters in the world, and are commercially very important. Understanding the knowledge of movements and habitat preferences of bigeye tuna enables improving the efficiency of the long-line fleet, aiding in resource management and sustainable use of fisheries resources.^[3] Bigeye tuna can be deep swimmer and highly mobile, which allows them have distinctive depth distribution and vertical movement

patterns. They remain in the uniformed temperature surface layer at night and can descend to greater than 500 m depth at dawn. While exhibiting non-associative behaviour, they descend below the thermocline to forage the small nektonic organisms of the deep scattering layer (DSL). [4,5] Bigeye tuna are caught mainly by long-line, especially for adult fish. The efficiency of long-line gear differs depending on the depth of hooks and their relationship with the depth of fish.

Bigeye tuna prefers to feed at the upper of the thermocline and below. When the vertical depth of the thermocline is shallow, the active range of the bigeye tuna is smaller, the vertical motion is limited, and the distribution of bigeye tuna is relatively concentrated, making them more catchable, and the amount of catch increases. The vertical pattern of water temperature, especially thermocline distribution play a key factor in the fishing grounds distribution of bigeye tuna. So identifying the vertical distribution of bigeye tuna improves our understanding of long-line catches and provides critical information for fisheries management and resource conservation.

In recent years, with the development of technology, the methods of fishery have greatly updated. So the effort of bigeye tuna catching becomes much greater than before (Fig 4). During the period of 2006-2008, there is already an evidence showing that the increasing of catching efforts lead to a less harvest in fishery. In addition, the CPUE of bigeye tuna also gradually decreases, we assumed one initial point (blue point on Fig 5) as the original stock of bigeye tuna in WCPO area, from the graph we can infer that during the period of 2010-2015 (res points on Fig 5), bigeye tuna is potentially facing the danger of overfishing. In order to protecting them from overfishing, we need to know more about their habitats and make some efficient policies.

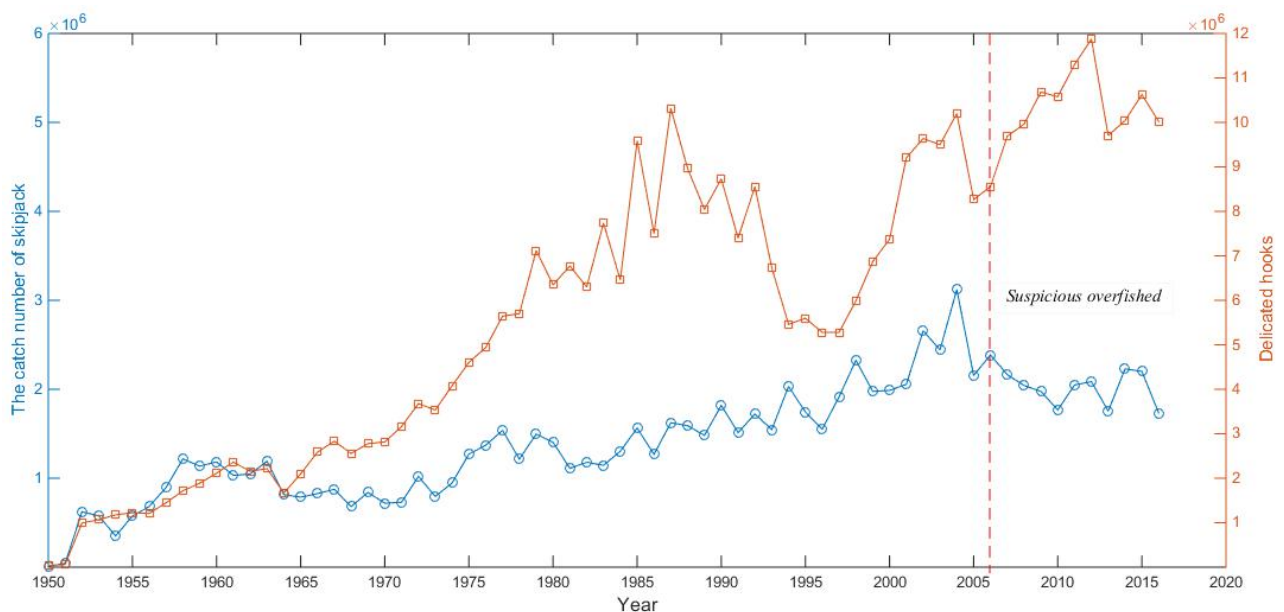


Fig 4 Change of used hooks and number of fishes catching (1950-2016)

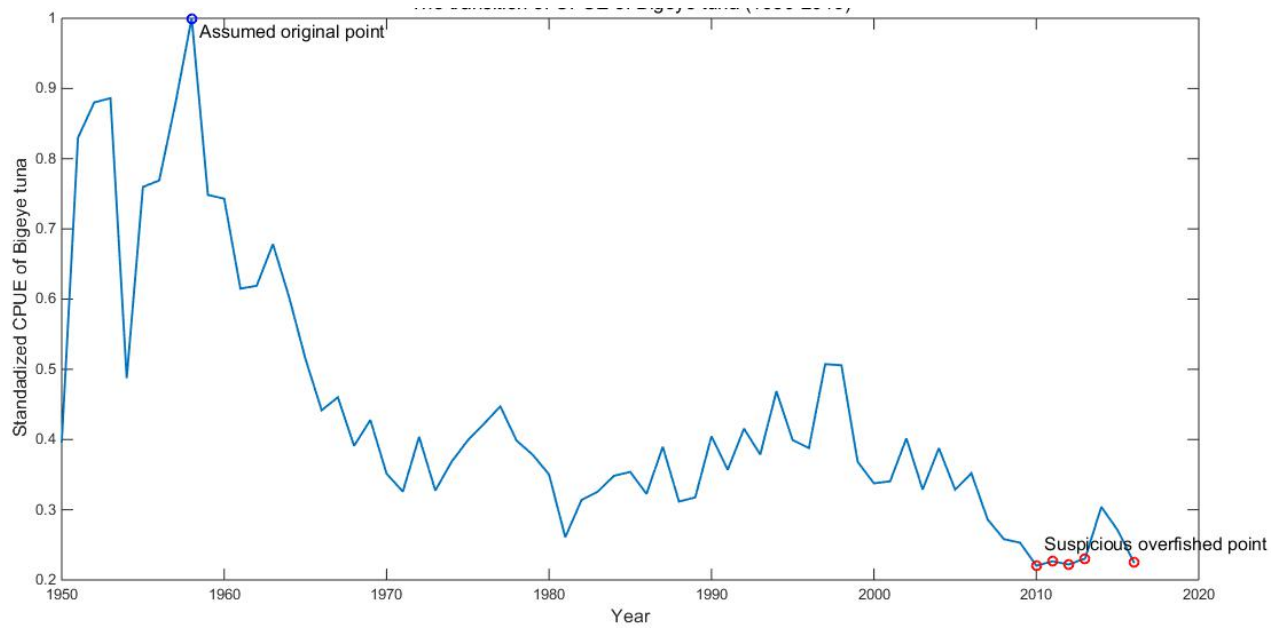


Fig 5 Transition of CPUE of bigeye fish tuna (1950-2016), showing the series of annual CPUE compare to the assumed original point, red points indicate potentially overfished

3.2 Data access and research method

In this paper, two types of data set were used: fishery and Argo buoys' datasets. Bigeye tuna inhabit tropical and temperate waters in the world, particularly for the tuna long-line fishery in the tropical area. To meet the objectives of this study, the location (110°E-130°W, 50°S-50°N) was used as the study area. The catch and catching effort data were obtained from WCPFC, from 2004 to 2016. They include the number of hooks, fishing time, longitude and latitude, and catch of bigeye tuna. The spatial resolution is 5°×5° and the temporal resolution is a month. The use of 5° grids is a conventional method to deal with tuna catch data.

The Argo floats data to describes the ocean variability and have been used extensively in different oceanic studies in recent years, but as yet have been seldom used in pelagic fishery research.^[6] In this paper, We use Argo data (BOA_Argo global Marine Argo grid data sets) to calculate the depth of thermocline and match it with the fishery data. We used the profile temperature data of the "BOA_Argo" grid data from 2004 to 2016 for thermocline analysis. The data selected for this grid dataset is from the 1997-2018 global ocean (79.5°S-79.5°N, 180°W-180°E) provided by China Argo Real-Time Data Center (<http://www.argo.org.cn/>). And we found the optimum ranges of thermocline characteristics of the central fishing grounds and learned more about the tuna resource management in WCPO. This paper took the monthly and seasonly CPUE of long-line bigeye tuna and monthly thermocline characteristics values to investigate the relationship between the fishing grounds distribution of bigeye tuna and thermocline.

The fishing data collected by the WCPFC is generally managed according to the grid. The location is recorded in latitude and longitude coordinates. The latitude and longitude coordinate system is divided into four quadrants by the equator (latitude 0°) and the Greenwich ruled longitude (longitude 0°). The gridding of data will

facilitate better regional management and analysis (Fig 6). The jurisdiction of the Western and Central Pacific Ocean Tuna Commission is in all of the four quadrants, so the downloaded data is also distributed among them.

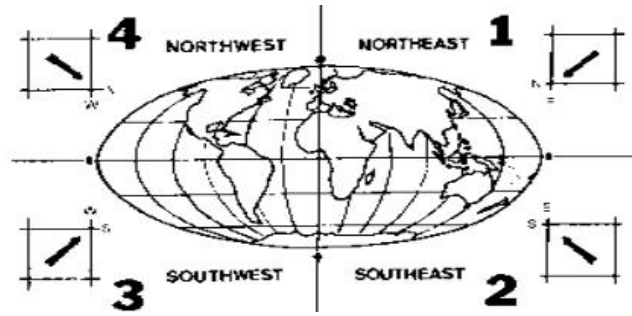


Fig 6 Divide of four quadrants

GRID AREA	Code
1° × 1°	5
5° × 5°	6
5° × 10°	1
10° × 20°	2
10° × 10°	3
20° × 20°	4

Table 1 Code of grid area

SIZE	QUADRANT	LATITUDE	LONGITUDE
6	1	50	140

Table 2 Example of an used grid

The downloaded data does not directly give the latitude and longitude used to programming, but gives a grid data, which requires us to obtain the latitude, longitude, grid and quadrant distribution information according to the data description. In this study, only the grid resolution of 5°×5° data was used. Example of data processing refers to Table 1, 2, as we can see, the value of grid is 61050014000 (Tables 1,2). The grid code is number 6, so the grid resolution is 5°×5°. The number 1 indicates that the quadrant of the data is in the first quadrant. The direction of the arrow in Figure 6 indicates the starting coordinate point of the grid. For the different quadrants, the starting position of the grid of the data is also different, so the data is classified and processed during processing. The number 50 indicates the latitude is 50°, the number 140 means the longitude is 140°, the latitude is represented by two digits, and the longitude is represented by three digits. If the processed data is not in the first quadrant, it should be noted that its latitude should be the data of the southern/western hemisphere, so it needs to be corrected to a negative value in the program according to the corresponding regulations.

After obtaining the latitude and longitude data, in order to facilitate the matching or superposition with the

Argo data, the center position of the 5°×5° grid is set as the center point of the fishing data. According to the fishing center point, the relevant data falling within the grid is matched in one piece as an observation unit. The downloaded fishery data also includes the year and month, the number of hooks, the operation fleet, and the type of work ect. We use the total catch fishtails and the total number of hooks to calculate the CPUE. The calculation formula for the unit fishing effort per CPU (CPUE, unit: fishtails / thousand hooks) per 5°× 5° statistical grid is:

$$CPUE_{(i,j)} = \frac{N_{fish(i,j)} \times 1000}{N_{hook(i,j)}}$$

The meaning of $CPUE_{(i,j)}$, $N_{fish(i,j)}$, $N_{hook(i,j)}$ is showing in the following: the monthly average CPUE of i-th longitude, j-th longitude grid, the total catch fishtails and the total number of hooks. This formula can eliminate the effects of different low-latitude and high-latitude grid sizes after projection. The CPUE was used as a relative abundance index of bigeye tuna. It was calculated as the number of individuals caught per 1000 hooks on a 5°×5° grid, and values averaged monthly for the data were spatially scattered and not evenly distributed.

To investigate monthly distribution patterns, we calculated the average CPUE for study years in all grids. Matlab software was used to create average CPUE distribution maps (Fig.7, Fig.8). We pooled monthly data and calculated the 4th quartile. We selected the 4th quartile (Q4=1.8) as the threshold. For this study, bigeye tuna CPUE were divided into four cases: (1) Cases with CPUE lower than 0.8 but greater than 0.8 – ‘low CPUEs’; (2) Cases with CPUE greater than 0.8 but lower than 1.2; (3) Cases with CPUE greater than 1.2 but lower than 1.8; and (4) Cases with CPUE greater than 1.8 – ‘high CPUEs’. We define the high CPUEs regions as hotspots or good fishing zones.

We then plotted the upper and lower boundaries’ temperature and depth, contoured on a spatial overlay map to characterize the hotspots, and to determine the relationship between bigeye tuna distribution and thermocline characteristics. We then matched both these ranges to determine the preferred ranges of the two environmental (the upper and lower boundaries temperature and depth) conditions. In this way, we can roughly describe the optimum condition that the bigeye tuna would stay, furthermore we can know more about the distribution of this specie, then we can make some precautions in different seasons to protect them.

3.3 Result analysis

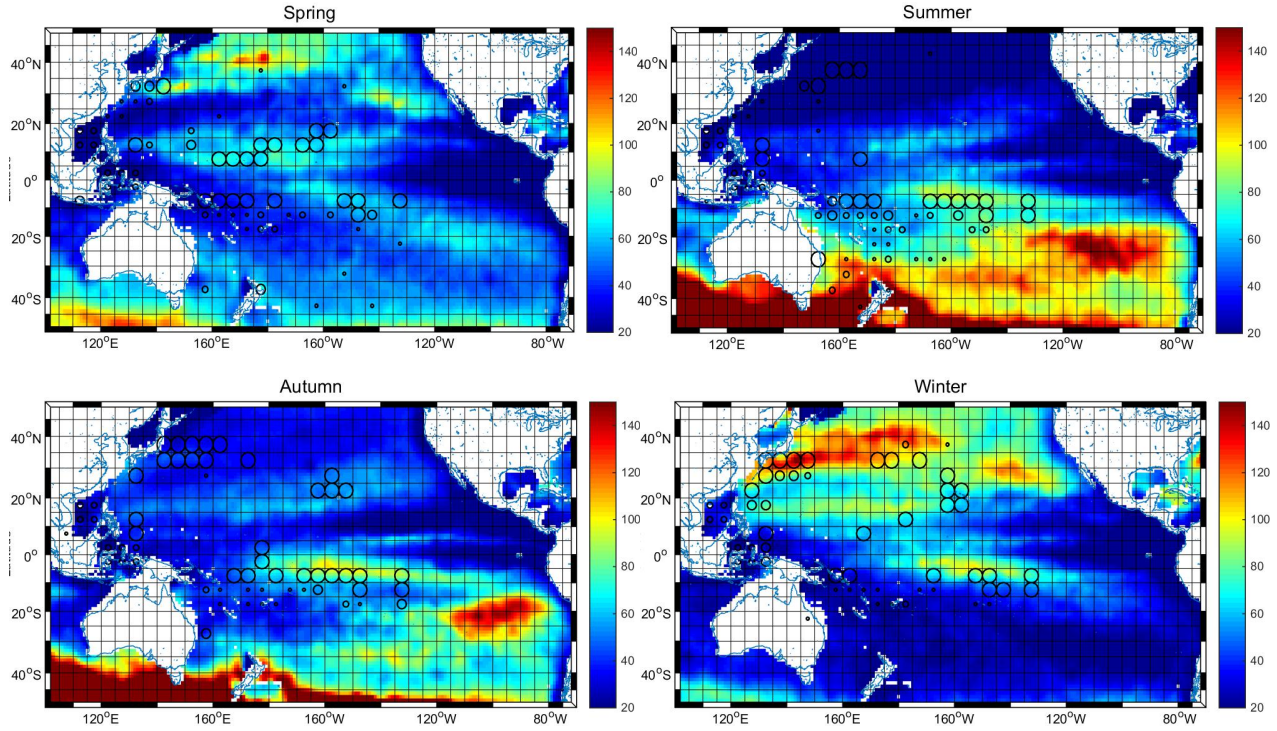


Fig 7a Upper thermocline boundaries depth, overlap bigeye fish CPUE distribution data (2004-2016), the size of circle presents the CPUE value in one grid unit (5°×5°)

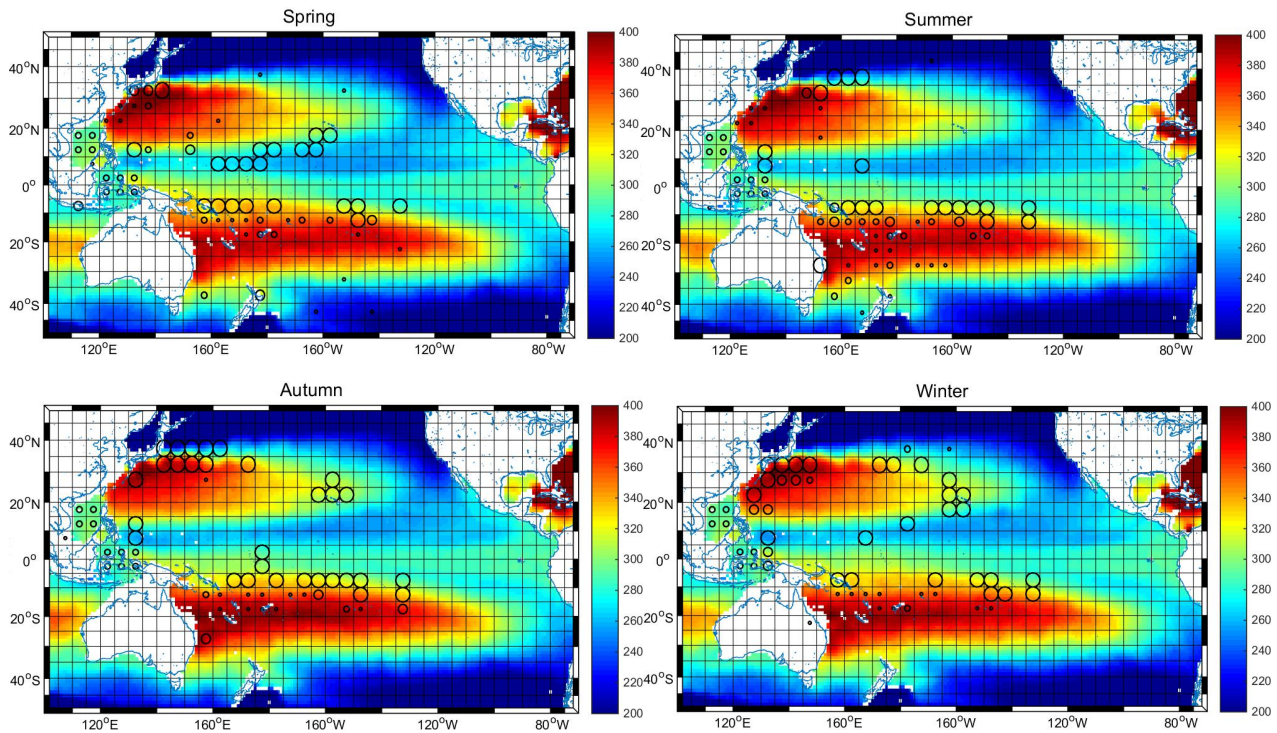


Fig 7b Lower thermocline boundaries depth, overlap bigeye fish CPUE distribution data (2004-2016), the size of circle presents the CPUE value in one grid unit (5°×5°)

Comparing Fig.7a and Fig.7b, it can be found that the upper layer of the thermocline is more obviously changes with seasons, while the change of the lower layer of the thermocline is relatively small, and much more stable. The occurrence of this phenomenon may be related to the distribution of the wind field. The upper layer of

the thermocline is susceptible to wind stress, while the lower layer is relatively less affected by wind stress, because of the frictional dissipation of seawater.

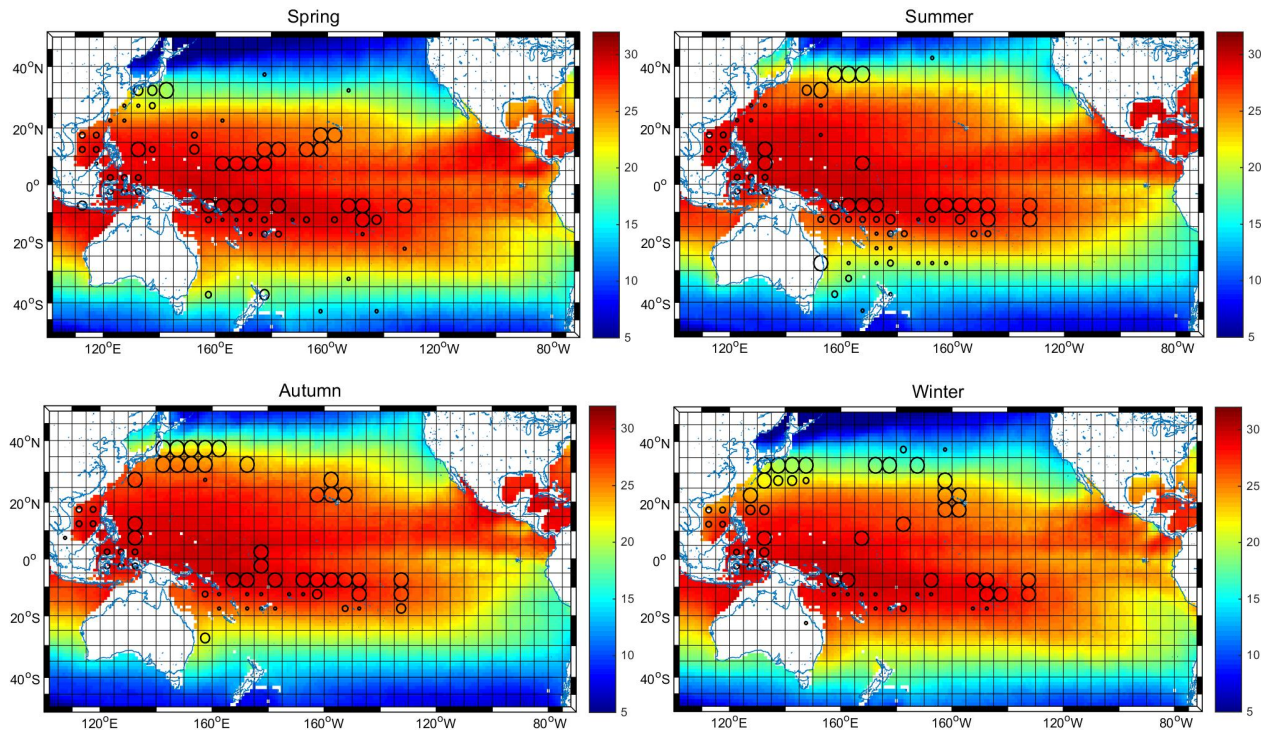


Fig 8a Upper thermocline boundaries temperature, overlap bigeye fish CPUE distribution data (2004-2016), the size of circle presents the CPUE value in one grid unit (5°×5°)

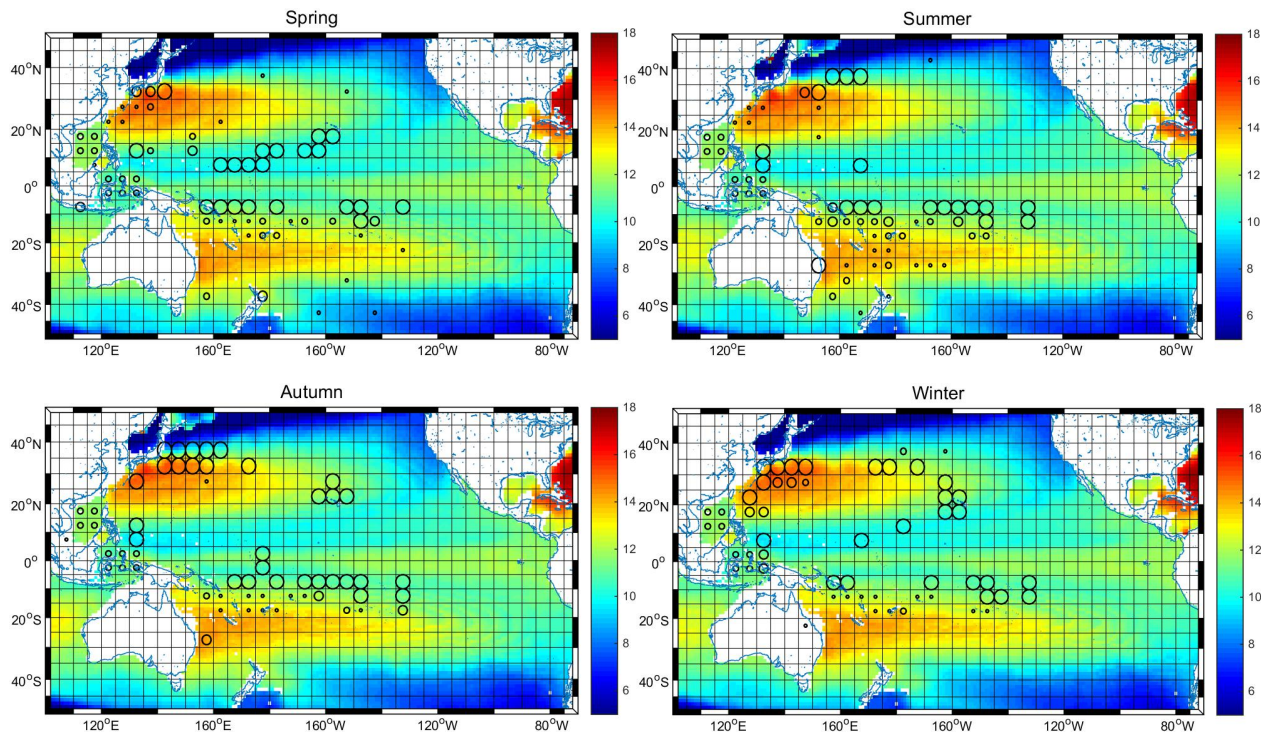
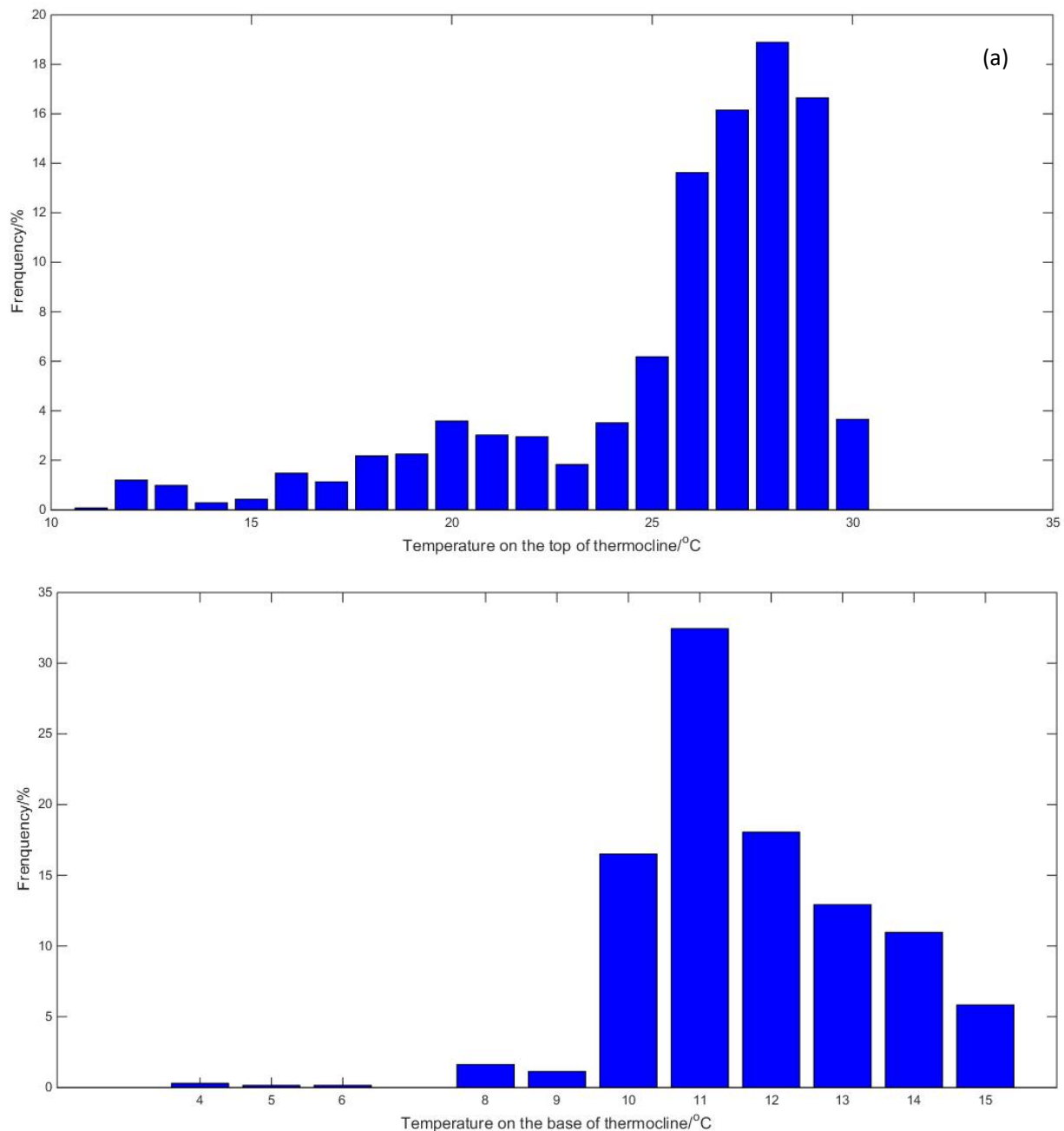


Fig 8b Lower thermocline boundaries temperature, overlap bigeye fish CPUE distribution data (2004-2016), the size of circle presents the CPUE value in one grid unit (5°×5°)

According to Fig. 8a, b, it can be seen that the temperature of the upper and lower boundaries of the thermocline is not obviously changed in each season, and it is relatively stable. It can be seen from the comparison

that the area with high catch of bigeye tuna (high CPUE), shows that the upper boundary temperature of the thermocline is relatively low, while the temperature of the lower boundary of the thermocline is moderate. The occurrence of this phenomenon is related to the optimal range of predation and habitat.

These graphs show the multi-year average seasonal distribution of the thermocline depth, so the ENSO phenomenon has little effect in this conclusion. It can be seen from the distribution of CPUE that the high value area of CPUE mainly exists in the area where the upper layer of the thermocline is shallow and the lower layer of the thermocline is medium depth. At the same time, although the lower layer of the thermocline in the offshore area of Japan has a large depth, the catch of bigeye tuna is relatively higher than other offshore place, but the specific reasons of this phenomenon are still unclear and need more physical and chemical parameters for more accurate conclusions.



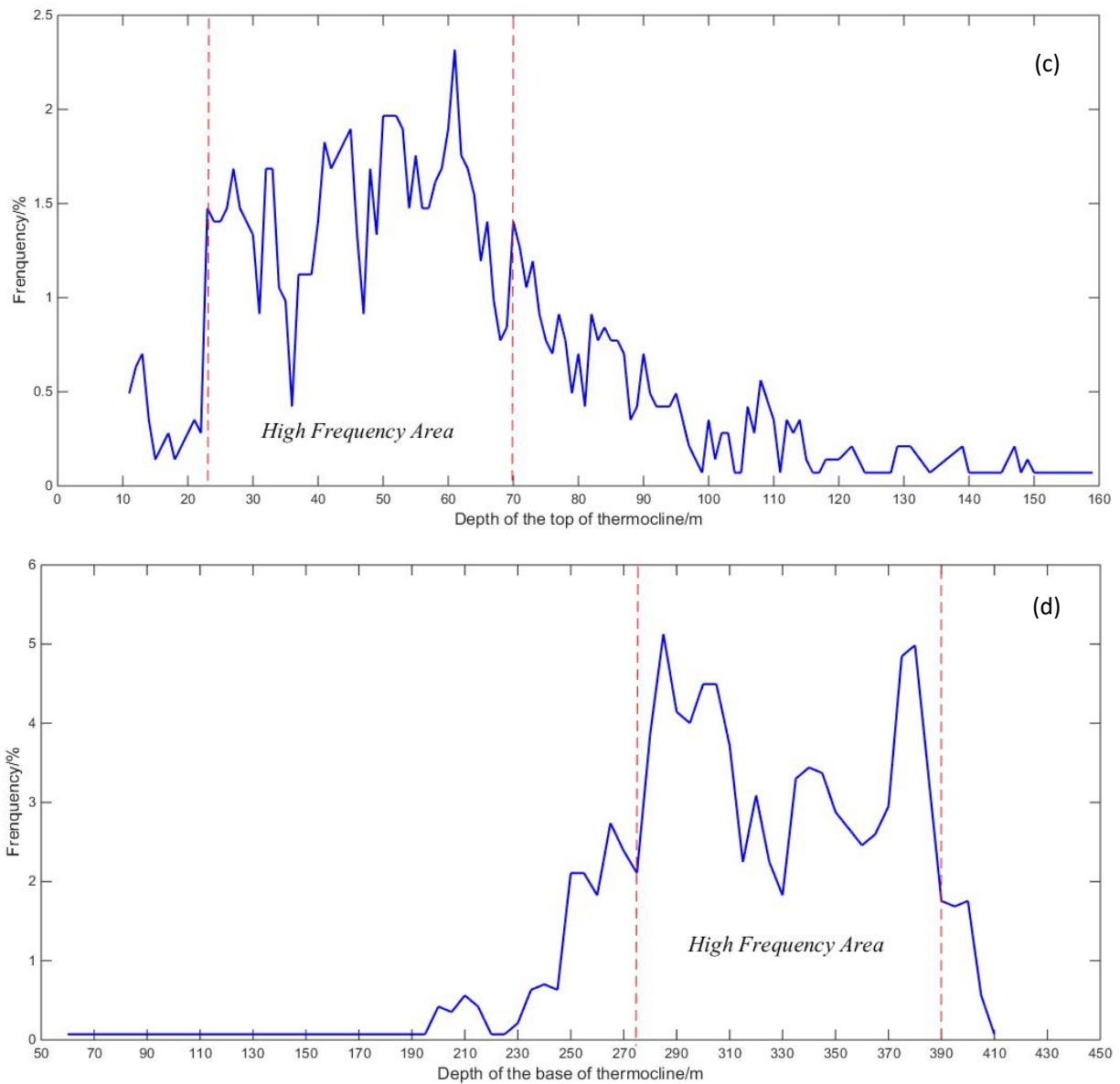


Fig 9 Relationship between thermocline variables and fishing frequency of high CPUE data for bigeye tuna CPUE during 2004-2016

As we can see in the picture (Fig 9 a,b,c,d), the frequency graph suggests that the central fishing grounds in the areas where the upper boundary temperature ranged from 15 to 30 °C (Fig. 9a), 80% of the highest CPUEs distributed between 25 and 30 °C and tended to be located at 28 °C. The lower boundary temperature were found in fishing grounds ranged from 8 to 15 °C (Fig. 9b), 94% of the highest CPUEs distributed between 10 and 15°C. The histogram of high catch rates suggests that fishing tended to be centered at 10~12 °C. The frequency of the upper boundary depth of high catch of bigeye tuna follows a Gaussian distribution. Distribution of high CPUEs in relation to the upper boundary depth ranged from 20 to 150m (Fig. 9c), 90.2% of the highest CPUEs distributed between 50 and 100m. The highest CPUEs in fishing grounds occurred in areas where the lower boundary depth ranged from 200 to 410 m (Fig. 9d), 77% of the highest CPUEs distributed 275~390 m.

3.4 Discussion

Bigeye tuna inhabit tropical and temperate waters in the world, and spawned in warm water. Obviously, the water temperature of bigeye tuna inhabit area influences the tuna spatial distribution directly. Acoustic telemetry and archival tags researches have proved that bigeye tuna displayed a distinct diurnal shift in diving behavior, generally diving at dawn to deeper, cooler waters and returning to shallower, warmer waters at dusk^[8]. The thermocline plays a key role in decision making about vertical habitat preferences of bigeye tuna and influences the efficiency of long-line gear.

Bigeye tuna remains in the shallow and warmer surface layer at night there by vertical distribution became narrow, which increases the vulnerability of tuna to surface fishing gears^[9]. Evans^[10] reported that there was a strong diurnal trend in bigeye catch rates, with catches from night sets generally higher than those from day sets. The optimum ranges for the upper boundary thermocline temperature and depth were between 26-29 °C (Fig 9a) and 50-100m (Fig 9c) in the paper, respectively. The results from archival tagging suggest that bigeye tuna vertical distributed between 24-26 °C during night in the Western Coral Sea and the Central North Pacific Ocean. The Figure 9a showed that CPUE is low in areas where the upper boundary temperature of thermocline was higher than 29 °C or less than 25 °C.

In recent years Argo data have been applied to a wide range of ocean areas, yet very little information is available on fisheries. In this paper, we used Argo profile data to reconstruct the subsurface environment field and to reveal the habitat of bigeye tuna. The results provide new insights into how oceanographic features influence the habitat of tropical pelagic fish and fisheries then exploit them by using a new tool (Argo profile data). Thus, the resolution of the subsurface environmental field may not compare well with the assimilation method, but it provides a convenient and effective way to investigate the habitat of tuna.^[11] However, the habitat of bigeye tuna is not defined by a single variable – temperature, depth at temperature, thermocline, oxygen, SSL distribution, and, undoubtedly, a host of other factors interact to define the habitat of bigeye tuna. Fortunately, most of the variables that may influence fish can be obtained by Argo data.

4. Conclusion

In this paper we found that the distribution of the abundance index CPUE (Catch Per Unit Effort) shifts longitudinally in concert with ENSO events distributed eastwards during El Niño phases, and westwards during La Niña phases, which shows an important relationship between the ENSO phenomena and fisheries.

The present study reveals the relationship between the fishing grounds and thermocline in the Western and Central Pacific Ocean. There are significant seasonal variations in the upper boundary temperature and depth of thermocline in the central fishing grounds and significantly influenced the temporal and spatial distribution of the

bigeye tuna population. The optimum ranges for the upper boundary thermocline temperature and depth and the lower boundary thermocline temperature and depth, were between 25- 30 °C, 50-100 m, 10-12 °C, 275-390 m, respectively.

Because of the limit of time and self-capacity, there are still some errors in this paper and in the computer programming. For the further study, we can focus more on other problems (for example: acidification of seawater, the distribution of nutrients and chlorophyll, the increasing of sea level, the climate change, ect.). After completing the basic parts, the research idea can also specifically consider the effects of meso-scale eddy (cold eddy, warm eddy) and upwelling on the migration of tuna fish, and also match the upper and lower boundary depths of the mixed layer with the temperature data and the CPUE data. We can do the parameters correlation analysis, the overall research results hope to be more conducive to understanding the impact of changes in marine environment on highly migratory fish.

5. Acknowledgement

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